

Problem Identification

⚠ The Microsleep Crisis on Malaysian Roads

Road safety remains a critical national concern in Malaysia, with fatigue-related accidents representing one of the most preventable yet persistent threats on our highways. According to the Malaysian Institute of Road Safety Research (MIROS, 2023), drowsy driving contributes to over **20% of fatal crashes annually**, second only to speeding as a cause of road fatalities. These accidents disproportionately spike during festive seasons when millions embark on long-distance journeys across the peninsula.

Target Users:

- **Primary:** Commercial drivers (express bus drivers, van operators, e-hailing drivers)
- **Secondary:** Long-distance travelers, night shift workers, and festive season drivers

Industry Focus:

Automotive safety technology and transportation sector, specifically addressing the gap in affordable driver monitoring systems for Malaysia's aging vehicle fleet.

✓ Project Goal

DashGUARD is an affordable, clip-on AI dashcam designed specifically for Malaysian conditions to detect microsleep precursors through advanced eye-tracking and facial analysis, providing alerts **2 seconds before loss of control occurs**.

Solution Strategy:

- 🔧 Utilize Single Board Computer processor for on-device real-time analysis of Microsleep from eyelid closure
- 🔔 Implement progressive 3-stage alert system (LED, Sound, App notification)
- 📞 Provide family safety net through automatic notifications to designated contacts
- 🔌 Ensure universal vehicle compatibility with plug-and-play installation

🌐 Global Impact Alignment



Aligning with UN Sustainable Development Goals

SDG 3: Good Health and Well-Being

Target 3.6: Reduce the number of global deaths and injuries from road traffic accidents.

DashGUARD Contribution:

- **Direct Impact:** Potential to prevent deaths and injuries annually in Malaysia through fatigue-related accident reduction.
- **Target Achievement:** Addresses Malaysia's plateaued road safety improvements by providing proactive prevention technology.
- **Health Outcome:** Reduces physical trauma, psychological impact on families, and long-term disability from preventable crashes.
- **Alignment:** Malaysia's Road Safety Plan 2022-2030 prioritizes human behavior improvement through technology.

Benchmarking Analysis

Comparison between DASHGUARD and existing market solutions.

USER NEEDS / PRODUCTS	DashGUARD	Toyota DVR	Tesla Active Safety	70mal T800	Garmin Live	VanTrue E360
1. PREVENT ACCIDENTS BEFORE LOSS OF CONTROL						
Detection Accuracy	>= 60%	Unavailable	~80%	N/A	~75%	N/A
Warning Time	>= 2 seconds	N/A	2-3 seconds	N/A	After drift	N/A
Microsleep Detection	Detection through facial landmarking to count eye closure timing, blinking ratio and head position	No	Moderate (Eye tracking)	No	Basic ADAS	No
2. EASY INSTALLATION						
Vehicle Compatibility	>= 100% of sockets	Toyota only	Tesla only	95%	95%	95%
Installation Type	Plug-and-play USB/Lighter	Fully Integrated	Fully Integrated	Wiring required	Wiring required	Wiring required
Professional Installation	Not required	N/A	N/A	Not required	Not required	Not required
3. WAKE DRIVER WITHOUT PANIC						
Max Volume	<= 85dB	N/A	Fixed loud	N/A	Basic	N/A
Alert System	3-Stage (LED, Sound, App)	None	Loud + Visual	None	Basic alerts	None
Response Time	<= 2 seconds	N/A	Fast	N/A	Moderate	N/A
Progressive Alerts	Yes (Stage 1 on detection)	No	No	No	No	No

Research Validation & Scientific Foundation

DashGUARD's core detection logic is grounded in established scientific metrics for driver drowsiness. Our system is validated against the following key peer-reviewed research papers:



Real-time driver monitoring system with facial landmark-based eye closure detection and head pose recognition (Nature, 2023)



Facial Landmark-based Real-Time Driver Drowsiness Detection (IRJET)



PERCLOS: A Valid Psychophysiological Measure of Alertness As Assessed by Psychomotor Vigilance (JOV)



PERCLOS Threshold for Drowsiness Detection during Real Driving (ResearchGate)



Validation of PERCLOS as a Measure of Alertness (NTL Repository)

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